

Tree formats and vocabulary

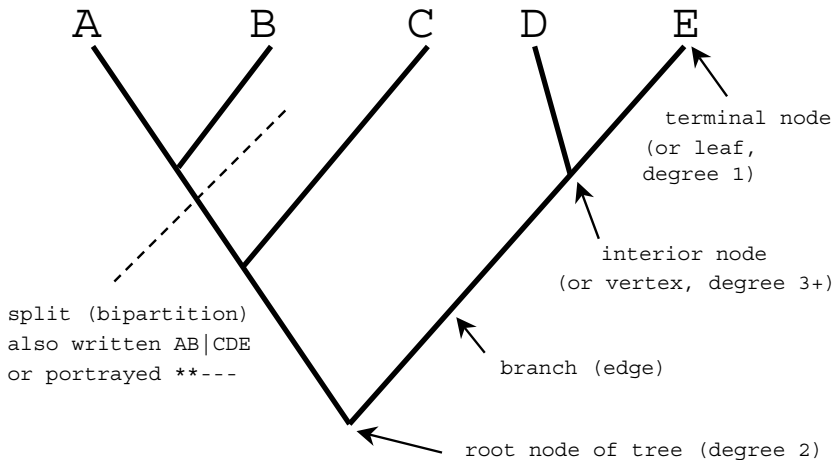
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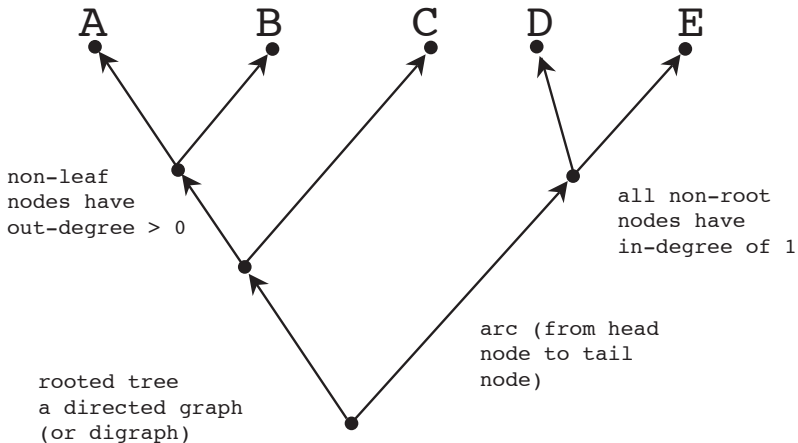
(With thanks to Mark Holder, Paul Lewis, Joe Felsenstein, and David Hillis for slides)

Phylogenies describe shared ancestry
and
inform our understanding of evolutionary processes

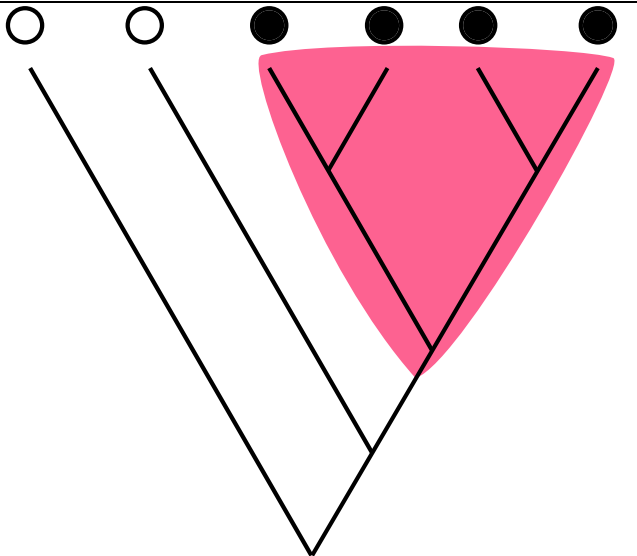
Tree terminology



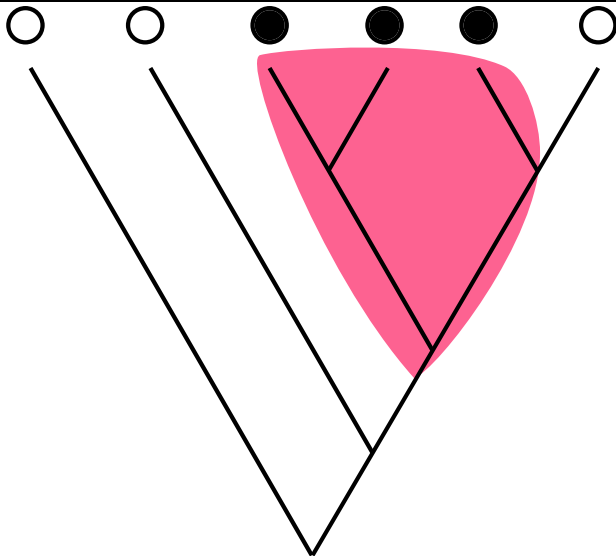
Rooted tree terminology

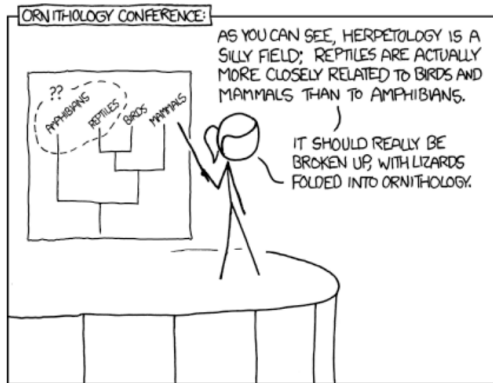


Monophyletic groups (“clades”): the basis of phylogenetic classification



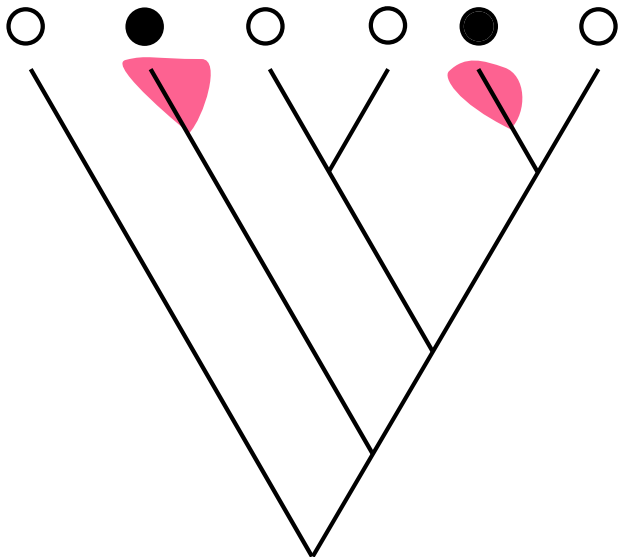
Paraphyletic groups: error of omitting some species



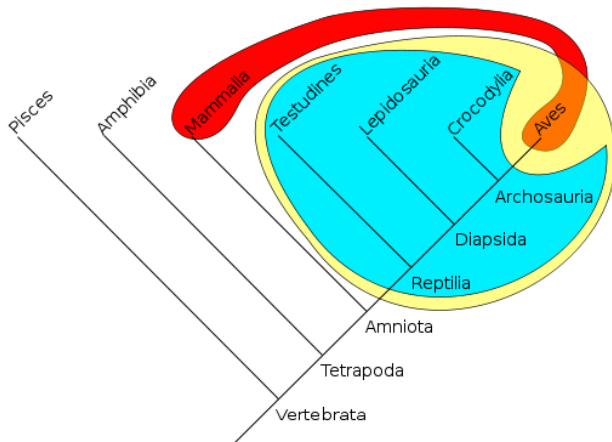


<https://xkcd.com/867/>

Polyphyletic groups: error of grouping “unrelated” species



- Monophyly
- Paraphyly
- Polyphyly

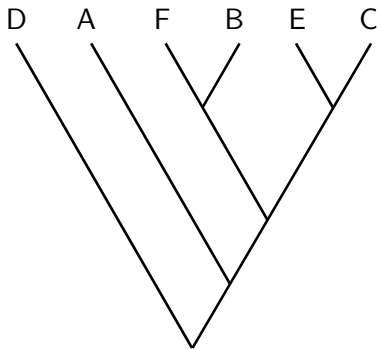
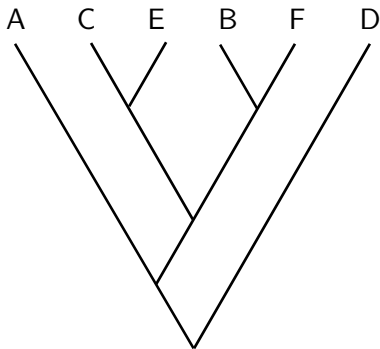


from wikipedia

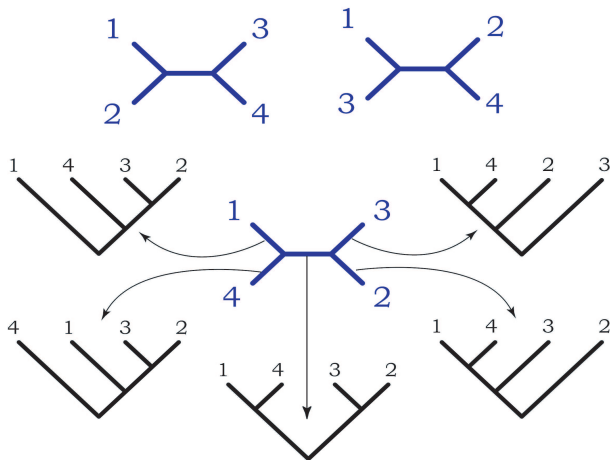
more terms:

- ▶ sister taxa: taxa or monophyletic groups which share a most recent common ancestor
- ▶ outgroup: taxon that is determined *a priori* to be sister to all other taxa in the analysis. Used for rooting tree

Branch rotation does not matter



Rooted vs unrooted trees



Splits

- ▶ It is useful to think of unrooted trees in terms of 'splits'
- ▶ Each branch in an unrooted tree splits the taxa into two groups.
- ▶ Membership in those groups can be denoted by ** vs ..
- ▶ e.g. a split between 1+2 and 3+4 can be summarized as

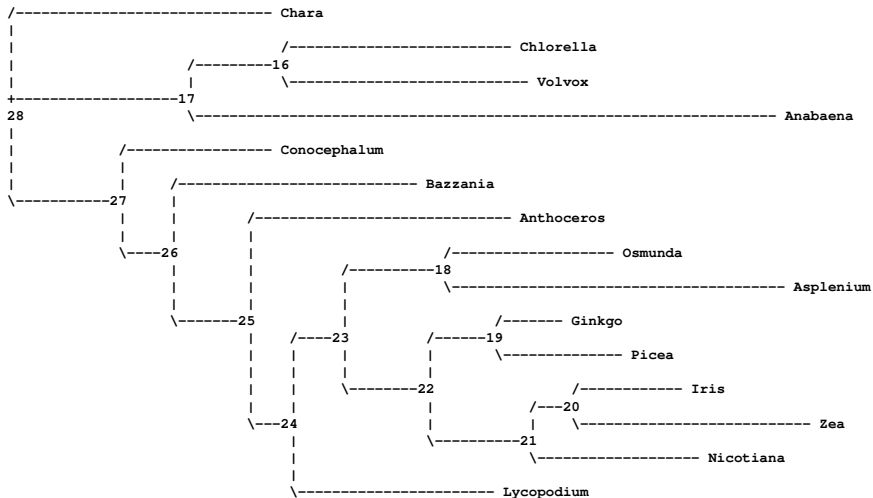
1234

** ..

or

12|34

Warning: software often displays unrooted trees like this:



Newick

- ▶ Parenthetical tree format
- ▶ Rooted vs. unrooted trees are not differentiated
- ▶ Some programs interpret polytomy at root as 'unrooted'
- ▶ Branches and nodes not well differentiated
- ▶ A name can contain any characters except blanks, colons, semicolons, parentheses, and square brackets

`((A,B)C);`

Nexus

- ▶ Starts with `#nexus`
- ▶ Can contain blocks of alignments, trees, commands, and more!
- ▶ Blocks between 'begin' and 'end'
- ▶ Trees in Newick format, prepended with [&U] unrooted or [&R] rooted

Nexus

```
#nexus
...
begin taxa;
  dimensions ntax=5;
  taxlabels
    Giardia
    Thermus
    Deinococcus
    Sulfolobus
    Haobacterium
;
end;
```

```
#nexus
...
begin data;
  dimensions ntax=5 nchar=54;
  format datatype=dna missing=? gap=-;
  matrix
    Ephedra      TTAAGCCATGCATGTCTAAGTATGAACTAATTCCAAACGGTGAAACTGCGGATG
    Gnetum        TTAAGCCATGCATGTCTATGTACGAACTAATC - AGAACGGTGAAACTGCGGATG
    Welwitschia   TTAAGCCATGCACGTGTGAAGTATGAACTAGTC - GAAACGGTGAAACTGCGGATG
    Ginkgo        TTAAGCCATGCATGTGTGAAGTATGAACTCTTTACAGACTGTGAAACTGCGAATG
    Pinus         TTAAGCCATGCATGTCTAAGTATGAACTAATTGCAGACTGTGAAACTGCGGATG
    [ ----+--10|----+--20|----+--30|----+--40|----+--50|----]
;
end;
```

http://hydrodictyon.eeb.uconn.edu/eebedia/index.php/Phylogenetics:_NEXUS_Format

Nexus

```
#nexus
...
begin trees;
  translate
    1 Ephedra,
    2 Gnetum,
    3 Welwitschia,
    4 Ginkgo,
    5 Pinus
  ;
  tree one = [&U] (1,2,(3,(4,5)));
  tree two = [&U] (1,3,(5,(2,4)));
end;
```

```
#nexus
...
begin sets;
  charset trnL_intron = 562-4226;
  taxset gnetales = Ephedra Gnetum Welwitschia;
end;
```

http://hydrodictyon.eeb.uconn.edu/eebedia/index.php/Phylogenetics:_NEXUS_Format

NeXML

- ▶ Phylogenetic data as XML
- ▶ Can capture all information from Nexus
- ▶ Full semantic annotation
- ▶ Easily extensible

NeXML

Computer readable, but not very human readable

```
<otu about="#otu99" id="otu99" label="Parupeneus barberinoides">
  <meta datatype="xsd:string" property="ot:originalLabel" xsi:type="nex:LiteralMeta">Parupeneus
  <meta datatype="xsd:int" property="ot:ottId" xsi:type="nex:LiteralMeta">758968</meta>
  <meta datatype="xsd:string" property="ot:ottTaxonName" xsi:type="nex:LiteralMeta">Parupeneus b
</otu>
</otus>
<trees about="#trees1" id="trees1" otus="otus1">
  <tree about="#tree1" id="tree1" label="Untitled (tree1)" xsi:type="nex:FloatTree">
    <meta datatype="xsd:string" property="ot:branchLengthDescription" xsi:type="nex:LiteralMeta"/>
    <meta datatype="xsd:string" property="ot:branchLengthMode" xsi:type="nex:LiteralMeta">ot:undef
    <meta datatype="xsd:string" property="ot:curatedType" xsi:type="nex:LiteralMeta">Bayesian infe
    <meta datatype="xsd:string" property="ot:ingroupClade" xsi:type="nex:LiteralMeta">node2</meta>
    <meta datatype="xsd:string" property="ot:nodeLabelMode" xsi:type="nex:LiteralMeta"/>
    <meta datatype="xsd:string" property="ot:nodeLabelTimeUnit" xsi:type="nex:LiteralMeta"/>
    <meta datatype="xsd:string" property="ot:outgroupEdge" xsi:type="nex:LiteralMeta"/>
    <meta datatype="xsd:string" property="ot:specifiedRoot" xsi:type="nex:LiteralMeta">node1</meta>
    <meta datatype="xsd:boolean" property="ot:unrootedTree" xsi:type="nex:LiteralMeta">false</meta>
    <node about="#node1" id="node1" root="true"/>
    <node about="#node2" id="node2"/>
    <node about="#node144" id="node144"/>
    <node about="#node145" id="node145"/>
    <node about="#node146" id="node146"/>
    <node about="#node147" id="node147"/>
    <node about="#node148" id="node148"/>
    <node about="#node149" id="node149"/>
    <node about="#node150" id="node150"/>
    <node about="#node151" id="node151"/>
    <node about="#node152" id="node152"/>
    <node about="#node153" id="node153"/>
    <node about="#node154" id="node154"/>
    <node about="#node155" id="node155" otu="otu72">
      <meta datatype="xsd:boolean" property="ot:isLeaf" xsi:type="nex:LiteralMeta">true</meta>
    </node>
    <node about="#node156" id="node156" otu="otu73">
      <meta datatype="xsd:boolean" property="ot:isLeaf" xsi:type="nex:LiteralMeta">true</meta>
    </node>
    <node about="#node157" id="node157" otu="otu74">
      <meta datatype="xsd:boolean" property="ot:isLeaf" xsi:type="nex:LiteralMeta">true</meta>
    </node>
    <node about="#node158" id="node158"/>
    <node about="#node159" id="node159" otu="otu75">
      <meta datatype="xsd:boolean" property="ot:isLeaf" xsi:type="nex:LiteralMeta">true</meta>
    </node>
    <node about="#node160" id="node160" otu="otu76">
```

Phylip (sequence data format)

- ▶ First line must be two integers: <number of taxa> <number of sites>
- ▶ Sequence ID followed by spaces up to 10 char.
- ▶ No duplicate names
- ▶ Relaxed phylip up to 250 characters followed by a space

```
      5      42
Turkey      AAGCTNGGGC ATTCAGGGT GAGCCCGGGC AATACAGGGT AT
Salmo gairAAGCCTTGGC AGTGCAGGGT GAGCCGTGGC CGGGCACGGT AT
H. SapiensACCGGTTGGC CGTTCAGGGT ACAGGTTGGC CGTTCAGGGT AA
Chimp       AAACCCTTGC CGTTACGCTT AAACCGAGGC CGGGACACTC AT
Gorilla     AAACCCTTGC CGGTACGCTT AAACCATTGC CGGTACGCTT AA
```

Phylip interleaved

```
5      42
Turkey   AAGCTNGGGC ATTCAGGGT
Salmo gairAAGCCTTGGC AGTGCAGGGT
H. SapiensACCGGTTGGC CGTTCAGGGT
Chimp     AAACCCTTGC CGTTACGCTT
Gorilla   AAACCCTTGC CGGTACGCTT

GAGCCCGGGC AATACAGGGT AT
GAGCCGTGGC CGGGCACGGT AT
ACAGGTTGGC CGTTCAGGGT AA
AAACCGAGGC CGGGACACTC AT
AAACCATTGC CGGTACGCTT AA
```

Phylip sequential

```
5      42
Turkey   AAGCTNGGGC ATTCAGGGT
GAGCCCGGGC AATACAGGGT AT
Salmo gairAAGCCTTGGC AGTGCAGGGT
GAGCCGTGGC CGGGCACGGT AT
H. SapiensACCGGTTGGC CGTTCAGGGT
ACAGGTTGGC CGTTCAGGGT AA
Chimp     AAACCCTTGC CGTTACGCTT
AAACCGAGGC CGGGACACTC AT
Gorilla   AAACCCTTGC CGGTACGCTT
AAACCATTGC CGGTACGCTT AA
```

Fasta (sequence data format)

- ▶ Description line before each sequence starts with (">") symbol in the first column

```
>AB000263 |acc=AB000263|descr=Homo sapiens mRNA for prepro cortistatin like peptide, complete cds.|len=368
ACAAGATGCCATTGTCCCCGGCCTCCTGCTGCTGCTCTCCGGGGCCACGGCCACCGCTGCCCTGCC
CCTGGAGGGTGGCCCCACCGGCCGAGACAGCGAGCATATGCAGGAAGCGGCAGGAATAAGGAAAAGCAGC
CTCCTGACTTTCTCGCTTGGTGGTTTGAGTGGACCTCCAGGCCAGTGCCGGGCCCCCTCATAAGGAGAGG
AAGCTCGGGAGGTGGCCAGGCGGCAGGAAGGCGCACCCCCCAGCAATCCGCGCGCCGGGACAGAATGCC
CTGCAGGAAC TTCTTGGAAGACCTTCTCCTCCTGCAAATAAAACCTCACCCATGAATGCTCACGCAAG
TTTAATTACAGACCTGAA
```

An Exercise: Writing and Viewing a Tree by Hand

- 1 Make sure you have:
 - ▶ A text editor installed.
 - ▶ *FigTree*, *PearTree* (or some other tree visualizer of choice).
- 2 Create a subdirectory for all our labs, e.g. `"$HOME/projects/GradPhylo"`.
- 3 Create a work subdirectory within this directory for this particular lab, e.g. `"lab-01"`.
- 4 Using a create a simple Newick tree file using a text editor, e.g. `"simple.tre"`.
- 5 This tree should reflect the real topological relationships of at least 4 species.
- 6 Visualize it in *FigTree*.
- 7 Export as a NEXUS file.
- 8 Open this second file in a text editor and edit the tip labels.
- 9 Visualize it again.

Create a newick tree file in your text editor with the content:

```
((C, (D,E)), (F,G), A), B);
```

Save it as 'example.tre'.

- 1 Draw the tree by hand
- 2 Write down all the splits in "AB|CD" format.
- 4 Load the tree in a tree viewer (e.g. phylo.io, figtree). Re-root the tree. What rootings make the following true? Which cannot be true?
 - a. A is more closely related to G than it is to C
 - b (C,D,E) is sister to (A,B,F,G)
 - c (C,D) is sister to (A,B,E,F,G)
 - d (C,D,E) is a paraphyletic group
 - e (C,D,E) is a monophyletic group
 - f (A,B,C) is a monophyletic group